An Elegant Expression for π in Terms of Complex Logarithms

Pearl Bipin Pulickal

April 4, 2025

Abstract

This paper presents an pedagogically valuable and elegant representation of the mathematical constant π derived from complex logarithms. Unlike traditional infinite series or trigonometric identities, the formula showcased offers a concise and direct connection between complex analysis and π, showcasing the deep interrelations between elementary functions and fundamental constants. However, the paper addresses important nuances such as the multi-valued nature of complex logarithms and the choice of the principal branch to ensure uniqueness.

1 Introduction

The constant π is one of the most important and well-known constants in mathematics, traditionally defined as the ratio of the circumference of a circle to its diameter. Over the centuries, numerous formulas have been proposed to express π, many of which involve infinite series or other advanced techniques such as continued fractions or trigonometric identities. While these representations have provided deep insights into the properties of π, they often involve infinite summations or iterative processes that can be complex and cumbersome.

In this paper, we introduce a compact expression for π, derived from the logarithm of the imaginary unit i. This formula is simple yet profound, as it connects π directly to the complex logarithmic function without relying on infinite series or trigonometric terms. We also explore the significance of branch cuts in the complex logarithm and discuss the necessary selection of the principal branch for uniqueness.

2 A Brief Primer on Euler's Formula and Complex Logarithms

To ensure accessibility for readers unfamiliar with the key concepts used in this paper, we provide a brief primer on Euler's formula and the complex logarithm.

2.1 Euler's Formula

Euler's formula is one of the most important and beautiful results in mathematics. It states that for any real number x, the exponential function with an imaginary exponent is related to trigonometric functions as follows:

$$e^{ix} = \cos(x) + i\sin(x).$$

This formula connects the exponential function to the sine and cosine functions, forming a bridge between algebraic and trigonometric analysis. When $x = \pi$, Euler's formula simplifies to the famous identity:

$$e^{i\pi} = -1.$$

This identity has deep implications in many areas of mathematics, including the study of complex numbers, Fourier analysis, and differential equations. It is often cited as one of the most elegant mathematical results, as it links five fundamental constants in a simple equation: e, i, π, 1, and 0.

2.2 The Complex Logarithm

The logarithm of a complex number extends the idea of the logarithm from real numbers to the complex plane. Just as the logarithm of a positive real number is the inverse of exponentiation, the complex logarithm is the inverse of the complex exponential function. For a complex number $z = re^{i\theta}$, where r is the modulus and θ is the argument (angle), the complex logarithm is defined as:

$$\log(z) = \log(r) + i(\theta + 2k\pi),$$

where k is any integer, representing the multi-valued nature of the logarithm. The argument θ is the angle formed by the complex number in the complex plane. The principal branch of the logarithm, typically denoted $\mathrm{Log}(z)$, restricts θ to the interval $(-\pi, \pi]$, ensuring a unique value for the logarithm.

For example, the logarithm of the imaginary unit i can be written as:

$$\log(i) = i\frac{\pi}{2}.$$

This result follows from the fact that i lies on the positive imaginary axis, and its argument is $\frac{\pi}{2}$. The principal value of the logarithm thus gives us the clean and simple result $\log(i) = i\frac{\pi}{2}$.

2.3 Multi-Valued Nature of the Logarithm

The complex logarithm is multi-valued because the argument of a complex number is not unique. In polar form, a complex number $z = re^{i\theta}$ can be written for any integer k as $z = re^{i(\theta+2k\pi)}$, which means that the argument θ can differ by integer multiples of 2π. As a result, the logarithm of a complex number is also multi-valued:

$$\log(z) = \log(r) + i(\theta + 2k\pi).$$

For the formula in this paper, we explicitly use the principal branch of the logarithm, where $k = 0$, to ensure that the formula for π is uniquely defined.

This multi-valued property of the logarithm is central to the study of complex functions and is an essential topic in complex analysis, especially when dealing with contour integration, branch cuts, and the behavior of complex functions in different regions of the complex plane.

3 Mathematical Derivation

We begin by recalling the polar form of the imaginary unit i, which is given by:

$$i = e^{i\frac{\pi}{2}}.$$

This relation follows from Euler's formula, which connects the exponential function to complex numbers. Taking the natural logarithm of both sides, we obtain:

$$\log(i) = i\frac{\pi}{2}.$$

Here, the logarithm $\log(i)$ is taken to mean the principal value, which ensures that the angle (or argument) is constrained to the interval $(-\pi, \pi]$, thus avoiding the multi-valued nature of the complex logarithm. The complex logarithm is multi-valued because for any integer k, we have:

$$\log(i) = i\left(\frac{\pi}{2} + 2k\pi\right).$$

To avoid ambiguity, we use the principal branch of the logarithm where $k = 0$, ensuring that the argument of i is precisely $\frac{\pi}{2}$.

Multiplying both sides by 2, we find:

$$2\log(i) = i\pi.$$

Finally, dividing both sides by i yields the simple yet elegant formula for π:

$$\frac{2\log(i)}{i} = \pi.$$

This equation expresses π directly in terms of the logarithmic function applied to the imaginary unit, showcasing the power and beauty of complex analysis.

4 Branch Cuts and Multi-Valued Nature of Complex Logarithms

The complex logarithm, as discussed, is inherently multi-valued due to the periodic nature of the argument. For any complex number z, the logarithm can be expressed as:

$$\log(z) = \log|z| + i(\arg(z) + 2k\pi),$$

where k is any integer and $\arg(z)$ is the argument of z. This means that without specifying the branch of the logarithm, the value of $\log(i)$ could take on multiple values depending on the integer k.

For the formula presented in this paper, we restrict ourselves to the principal branch, where the argument of i is $\frac{\pi}{2}$, thus ensuring a unique and consistent value. This is critical for the formula to hold as a well-defined expression for π.

5 Comparison with Known Results

The identity

$$\pi = -i \log(-1)$$

is well-known and can be derived from Euler's identity $e^{i\pi} = -1$. Our formula can be viewed as a specific case of this more general identity, where we have substituted $i = e^{i\frac{\pi}{2}}$ to obtain a direct connection between π and the logarithm of the imaginary unit. This formula does not introduce new concepts but provides a different perspective by utilizing the principal branch of the logarithm.

6 Significance and Implications

The proposed formula offers a straightforward and elegant way to represent π, bypassing the need for more complicated infinite series or summations. While infinite series such as the Leibniz series or the rapidly converging Chudnovsky formula are well-established in the computation of π, they involve iterative processes that can be difficult to handle in certain contexts. The formula presented here, by contrast, is a direct and closed-form expression, revealing a fundamental connection between π and complex logarithms.

Moreover, this result serves as a reminder of the deep interconnections between different areas of mathematics, particularly complex analysis and geometry. The simplicity of the formula highlights how seemingly abstract concepts—like the natural logarithm and the imaginary unit—can yield elegant expressions for widely known constants such as π.

While the formula does not provide a new computational tool for π (since it requires π itself for its evaluation), it is valuable as a pedagogical tool, helping students appreciate the connections between complex analysis and fundamental constants.

7 Applications in Complex Analysis, Number Theory, and Physics

The formula presented in this paper offers a fresh perspective on the interconnections between complex analysis and fundamental constants like π. While the formula itself does not introduce a new computational method for π, it serves as a pedagogical insight into several important areas of mathematics and physics.

7.1 Complex Analysis

In complex analysis, the natural logarithm of a complex number is a cornerstone concept. The formula $\frac{2\log(i)}{i} = \pi$ exemplifies how Euler's formula can be used to derive constants that play key roles in complex analysis. Euler's identity $e^{i\pi} = -1$ is often seen as one of the most elegant equations in mathematics, linking five fundamental constants: e, i, π, 1, and 0. The formula discussed in this paper can be seen as an extension of this identity, providing a new way to derive π using the properties of the complex logarithm.

Moreover, the multi-valued nature of the complex logarithm, which is discussed in this paper, has profound implications in understanding branch cuts and discontinuities in complex functions. These concepts are pivotal in contour integration, the study of analytic continuation, and the residue theorem, all of which are foundational to complex analysis.

7.2 Number Theory

In number theory, π frequently appears in the study of prime number distributions and analytic number theory, particularly in the Riemann zeta function. The formula for π derived here offers an alternative entry point to discuss the role of constants in such areas. By linking π to the complex logarithmic function, this result could serve as a bridge between algebraic and transcendental numbers, which are central themes in number theory.

Additionally, the deep connection between complex logarithms and modular forms could be explored, where identities involving π frequently arise in the study of these highly structured functions.

7.3 Physics

In physics, particularly in quantum mechanics, complex logarithms are used to describe oscillatory behavior, wave functions, and solutions to the Schrödinger equation. The formula for π can be applied in these contexts to illustrate the relationship between periodic functions and fundamental constants. The idea that π emerges from the logarithmic behavior of complex numbers ties into the broader theme of how mathematical constants govern the fundamental laws of nature.

Furthermore, the use of Euler's formula in physics often extends to understanding the behavior of complex systems, signal processing, and Fourier transforms. Thus, the formula for π derived here can be used to introduce these important concepts to students in physics and engineering.

7.4 Pedagogical Value

Overall, the formula for π provides a simple yet deep illustration of the power of complex analysis. It can be used in the classroom as a starting point for discussions on the history of π, the concept of complex logarithms, and the broader implications of Euler's formula. By presenting π as the result of a logarithmic relationship with the imaginary unit i, students can appreciate the interconnectedness of various mathematical concepts in a new light.

8 Conclusion

In this paper, we have presented a pedagogically valuable and elegant formula for the mathematical constant π that is derived from complex logarithms. This representation is both simple and profound, offering another perspective on a well-known constant without relying on infinite series or trigonometric identities. It emphasizes the beauty of mathematical relationships and the power of complex analysis in providing concise, elegant expressions for fundamental constants.

Future work could explore the potential applications of this formula in various fields, such as number theory, complex dynamics, and even quantum mechanics, where complex logarithms play a crucial role in understanding wave functions and oscillatory behavior. However, it is important to note that the formula is an identity rather than a computational method, and its value lies in deepening our understanding of mathematical connections rather than offering new practical algorithms.